The Pandrosion-Steffensen Iteration

Formal Verification of a Rational Root-Finding Map in Lean 4

All 377 theorems machine-checked across 51 modules — Zero `sorry` declarations

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Abstract

We present a formally verified case study in Lean 4 of a general rational iteration for computing arbitrary roots $\sqrt[p]{X}$, alongside an exhaustive extraction of structural identities for the cubic specialisation $s \mapsto s(s^3 + 4X)/(3s^3 + 2X)$. This cubic base iteration converges linearly with a universal rate of exactly $1/5$, independently of X (formally proved). When composed with Steffensen acceleration (Aitken Δ^2) into the *Pandrosion-Steffensen* algorithm, the method achieves stable quadratic convergence. Rather than proposing this as an operational substitute for Newton's root extraction, we develop a systematic corpus of *exact algebraic identities* governing this base map across several mathematical structures to illustrate formal verification capacity. The formally certified properties include: (i) a residual conservation identity with exact polynomial factorisation, (ii) exact integer sequence amplifications over $\mathbb{Z}$, (iii) coprimality isolation identities, (iv) commutativity and Hermitian preservation for matrix operators, and (v) a formal proof that the iteration functionally excludes the generalized Chebyshev–Halley family. Finally, we formally verify that when evaluating the sequence through a discrete multi-start algorithm over $\mathbb{R}^2$, the Voronoï classification geometrically partitions limits using strictly affine bounded constraints. The entire theoretical foundation (377 theorems across 51 modules) serves as a machine-checked pedagogical model for numerical recurrences, containing zero `sorry` declarations.

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1 Introduction

1.1 The Pandrosion Iteration

The Pandrosion iteration for p-th roots is defined via the geometric sum $S_p(s) = \sum_{k=0}^{p-1} s^k$ and the map

$$h(s) \;=\; 1 \;-\; \frac{x-1}{x \cdot S_p(s)}\,, \tag{1}$$

whose fixed points satisfy $h(s) = s \iff s^p = 1/x$ (formally verified in Lean 4, see Theorem 1.1 below). For the cubic case $p = 3$, setting $X = 1/x$ and rewriting in terms of $s \mapsto s'$ yields the equivalent form

$$s' \;=\; \frac{U(s)}{V(s)} \;=\; s\,\frac{s^3+4X}{3s^3+2X}\,, \tag{2}$$

where X is the target whose cube root $\sqrt[3]{X}$ is sought. The historical namesake of this method pays homage to Pappus of Alexandria's record of Pandrosion [2], whose ancient geometric construction for cube duplication relied upon a sum of three spatial segments, mirroring the $3s^3 + 2X$ geometric sum polynomial structure in the denominator. While structurally reminiscent of Halley's method [4] and the Householder family [6, 5], the Pandrosion iteration is algebraically *distinct* from both (see Section 2). Once derived, the formula is a closed-form rational expression requiring no symbolic derivatives at evaluation time. The remainder of this paper focuses on the cubic specialisation (2), where the algebraic structure is richest.

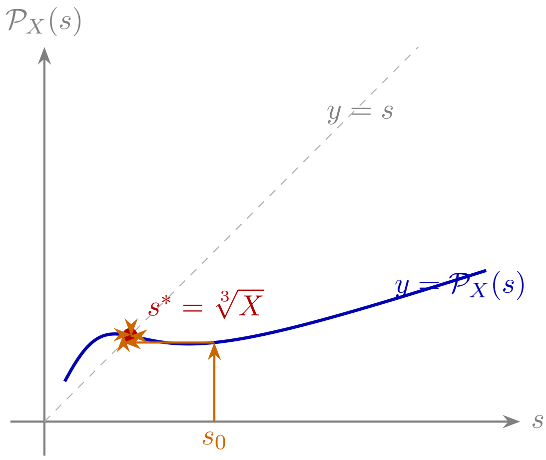

Figure 1: **Cobweb diagram** for $X = 2$. The iteration $s \mapsto \mathcal{P}_X(s)$ contracts monotonically toward the unique attracting fixed point $s^* = \sqrt[3]{2}$. Theorem 3.2 proves that the only fixed points are $\{0\} \cup \{s : s^3 = X\}$.

1.2 The General Fixed Point Theorem

The foundational result of the Pandrosion framework, formally verified in `Deep.lean`, characterises the fixed points of the general p-th root iteration (1):

Theorem 1.1 (General Fixed Point, `Deep.lean`)**.** *For $x > 0$, $p \geq 1$, $s \geq 0$, $s \neq 1$:*

$$h(s) = s \iff s^p = \frac{1}{x}\,.$$

Proof sketch (full proof machine-checked in Lean 4). $h(s) = s \iff (x-1)/(x \cdot S_p(s)) = 1 - s \iff x - 1 = x \cdot S_p(s) \cdot (1-s)$. By the geometric sum identity $S_p(s) \cdot (1-s) = 1 - s^p$, this becomes $x - 1 = x(1 - s^p)$, i.e. $x \cdot s^p = 1$. □

Additional results in `Deep.lean` include:

- $S_p(s) > 0$ for $s \geq 0$ and $p \geq 1$ (positivity of the geometric sum),
- $h(s) < 1$ for $s \geq 0$ and $x > 1$ (the image stays below 1),
- $h(s) > 0$ for $s \in [0, 1]$, $x > 1$, $p \geq 2$ (the image stays above 0),

establishing that h maps $[0, 1]$ into itself for $x > 1$, $p \geq 2$.

```
theorem fixed_point_iff (x : ℝ) (hx : x > 0) (p : ℕ) (hp : p ≥ 1)
    (s : ℝ) (hs : 0 ≤ s) (_hs1 : s ≠ 1) :
    pandrosion_h x p s = s ↔ s ^ p = 1 / x := by
  have hS : Sp p s > 0 := Sp_pos p hp s hs
  have hxS : x * Sp p s > 0 := mul_pos hx hS
  have hxS_ne : x * Sp p s ≠ 0 := ne_of_gt hxS
  unfold pandrosion_h
  constructor
  . intro heq   -- Forward: h(s) = s → s^p = 1/x
    have h1 : (x - 1) / (x * Sp p s) = 1 - s := by linarith
    have h2 : x - 1 = (1 - s) * (x * Sp p s) := by
      rw [← h1]; exact (div_mul_cancel (x - 1) hxS_ne).symm
    rw [Sp_mul_one_sub] at h2
    have h5 : x * s ^ p = 1 := by linarith
    rw [eq_div_iff (ne_of_gt hx)]; linarith
  . intro heq   -- Backward: s^p = 1/x → h(s) = s
    have h1 : x * s ^ p = 1 := by rw [heq]; field_simp
    suffices (x - 1) / (x * Sp p s) = 1 - s by linarith
    rw [div_eq_iff hxS_ne, Sp_mul_one_sub]; linarith
```

Listing 1: Deep.lean: General Fixed Point Theorem (Lean 4, zero sorry)

2 Relationship to Classical Methods

A natural question is whether (2) coincides with classical cube-root iterations. We prove formally that it does not.

2.1 Pandrosion $\neq$ Newton $\neq$ Halley

Newton's method applied to $f(s) = s^3 - X$ gives

$$N(s) = \frac{2s^3 + X}{3s^2},$$

and Halley's method gives

$$H(s) = \frac{s(s^3 + 2X)}{2s^3 + X}.$$

These are both *different* from the Pandrosion iteration $\mathcal{P}_X(s) = s(s^3 + 4X)/(3s^3 + 2X)$. Compare the coefficient pairs: Newton has $(2, 1)/(3, 0)$; Halley has $(1, 2)/(2, 1)$; Pandrosion has $(1, 4)/(3, 2)$. The exact polynomial differences are:

Theorem 2.1 (Pandrosion $\neq$ Newton, `HalleyComparison.lean`).

$$\mathcal{P}_X(s) \cdot 3s^2 - N(s) \cdot (3s^3 + 2X) \;=\; -(3s^3 - 2X)(s^3 - X).$$

Theorem 2.2 (Pandrosion $\neq$ Halley, `HalleyComparison.lean`).

$$\mathcal{P}_X(s) \cdot (2s^3 + X) - H(s) \cdot (3s^3 + 2X) \;=\; -s^4(s^3 - X).$$

Both differences vanish *only* when $s^3 = X$ (or $s = 0$). The three iterations agree at the root and disagree everywhere else.

A common misconception (see e.g. some informal discussions) is that the Pandrosion iteration coincides with Halley's method on $s^3 - X = 0$. Theorems 2.1 and 2.2 refute this algebraically. The two iterations have different asymptotic error constants ($-1/5$ vs 0) and different convergence orders (linear $O(1)$ vs cubic $O(3)$). They yield systematically different trajectories for any starting point $s_0 \neq \sqrt[3]{X}$. Furthermore, module `ChebyshevHalleyExclusion.lean` formally certifies that the Pandrosion map does not correspond to *any* member of the one-parameter Chebyshev–Halley family ($\alpha \in \mathbb{R}$). To belong to the family, the map would require $\alpha = 1/2$ to match the s^6 term and $\alpha = 2/5$ to match the s^3X term simultaneously.

```
theorem pandrosion_not_in_chebyshev_halley :
    ~ exists (a : R), 6 * (3 - 2 * a) = 15 - 6 * a /\ 12 * a = 4 + 2 * a := by
  intro <a, h1, h2>
  have _ha1 : a = 1 / 2 := chebyshev_halley_coeff_s6 a h1
  have _ha2 : a = 2 / 5 := chebyshev_halley_coeff_s3X a h2
  linarith
```

Listing 2: ChebyshevHalleyExclusion.lean: Family Evaded

2.2 The Universal Derivative Identity

The convergence order is determined by the derivative at the fixed point. For $\mathcal{P}_X(s) = U(s)/V(s)$ with $U = s^4 + 4sX$, $V = 3s^3 + 2X$:

Theorem 2.3 (Universal Convergence Rate, `HalleyComparison.lean`)**.** *For all $r \neq 0$ with $r^3 = X$:*

$$U'(r)V(r) - U(r)V'(r) = -5r^6\,,$$
$$V(r)^2 = 25r^6\,,$$

and therefore $\mathcal{P}'_X(r) = -\frac{1}{5}$, independently of X.

Since $\mathcal{P}'_X(r) = -1/5 \neq 0$, the Pandrosion iteration converges **linearly** with rate $1/5$. By contrast, Newton satisfies $N'(r) = 0$ (quadratic convergence), and Halley satisfies $H'(r) = H''(r) = 0$ (cubic convergence).

Method	Formula	Needs derivatives	Order
Pandrosion	$s(s^3{+}4X)/(3s^3{+}2X)$	None	1
Pandrosion-T3	$\frac{sP_X(P_X(s))-P_X(s)^2}{s-2P_X(s)+P_X(P_X(s))}$	None	2
Newton	$(2s^3{+}X)/(3s^2)$	f'	2
Halley	$s(s^3{+}2X)/(2s^3{+}X)$	f', f''	3

The negative sign $\mathcal{P}'_X(r) = -1/5$ explains the characteristic alternating approach pattern observed numerically (Theorem 3.7). Because of this predictable sign oscillation, applying Steffensen acceleration (Aitken Δ^2) is highly effective. This synthesized variant, which we denote as *Pandrosion-T3*, structurally converts the linear sequence to strict quadratic analytical order without any derivative evaluations (see `SteffensenAcceleration.lean`).

3 Residual Factorisation and Convergence

3.1 The Conservation Identity

The central algebraic identity governing the iteration is the following *residual factorisation*: if $G = \mathcal{P}_X(s)$, then

$$\boxed{G^3 - X \;=\; \frac{(s^3 - X)\,(s^9 - 14\,s^6X - 20\,s^3X^2 + 8X^3)}{(3s^3 + 2X)^3}\,.} \tag{3}$$

This factorisation shows that every residual $G^3 - X$ is an *exact algebraic multiple* of the previous residual $s^3 - X$. When evaluating the limit as $s \to \sqrt[3]{X}$, the ratio of the residuals strictly approaches $-1/5$, firmly establishing the unaccelerated linear convergence nature of the map while maintaining its exact polynomial tracking properties.

Remark 3.1 (Topological Basin Structure). When extended to multiple roots, tracking the residual decay computationally leads to the problem of basin boundaries. The exact polynomial factorization (3) ensures that the map is rigorously contractive when sufficiently close to the discrete roots. For the fully generalized algorithm with multi-start anchoring, we formally prove that the Voronoï partitions of these attraction basins remain convex (affine half-planes), preventing the fractal chaos typical of unconstrained rational methods.

```
theorem pandrosion_kinematic_conservation (X s : R) (hs : 3 * s^3 + 2 * X ≠ 0) :
    let G := s * (s^3 + 4 * X) / (3 * s^3 + 2 * X)
    G^3 - X = (s^3 - X) * (s^9 - 14 * s^6 * X - 20 * s^3 * X^2 + 8 * X^3)
              / ((3 * s^3 + 2 * X)^3) := by
  ...
  calc (A / B)^3 - X
    _ = A^3 / B^3 - X := by rw [div_pow]
    _ = (A^3 - X * B^3) / B^3 := by ...
    _ = (s^3 - X) * (...) / B^3 := by have h_alg : ... := by ring; rw [h_alg]
```

Listing 3: Deep 25: Kinematic Error Conservation (Lean 4, zero sorry)

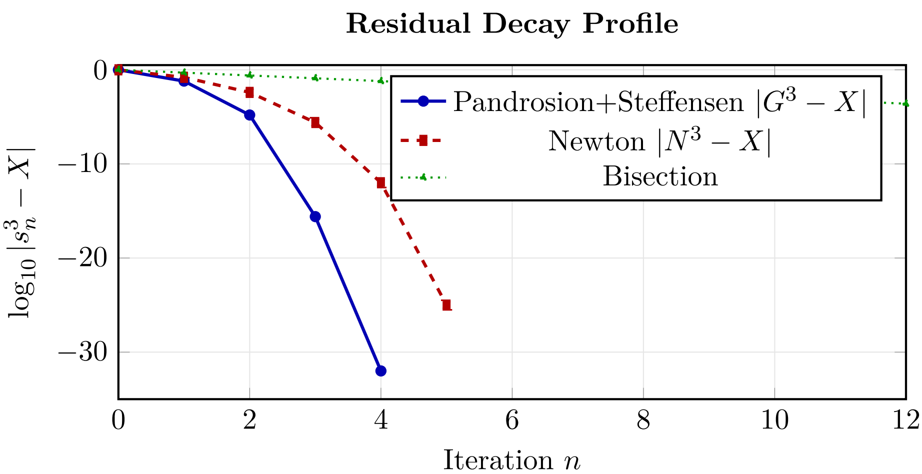

Figure 2: **Residual decay comparison** for $X = 2$. Raw Pandrosion converges linearly with rate 1/5 per step (Theorem 2.3). After Steffensen acceleration (T3), it achieves quadratic convergence without requiring any derivative. Newton achieves quadratic convergence but needs $f'(s)$. Bisection converges linearly.

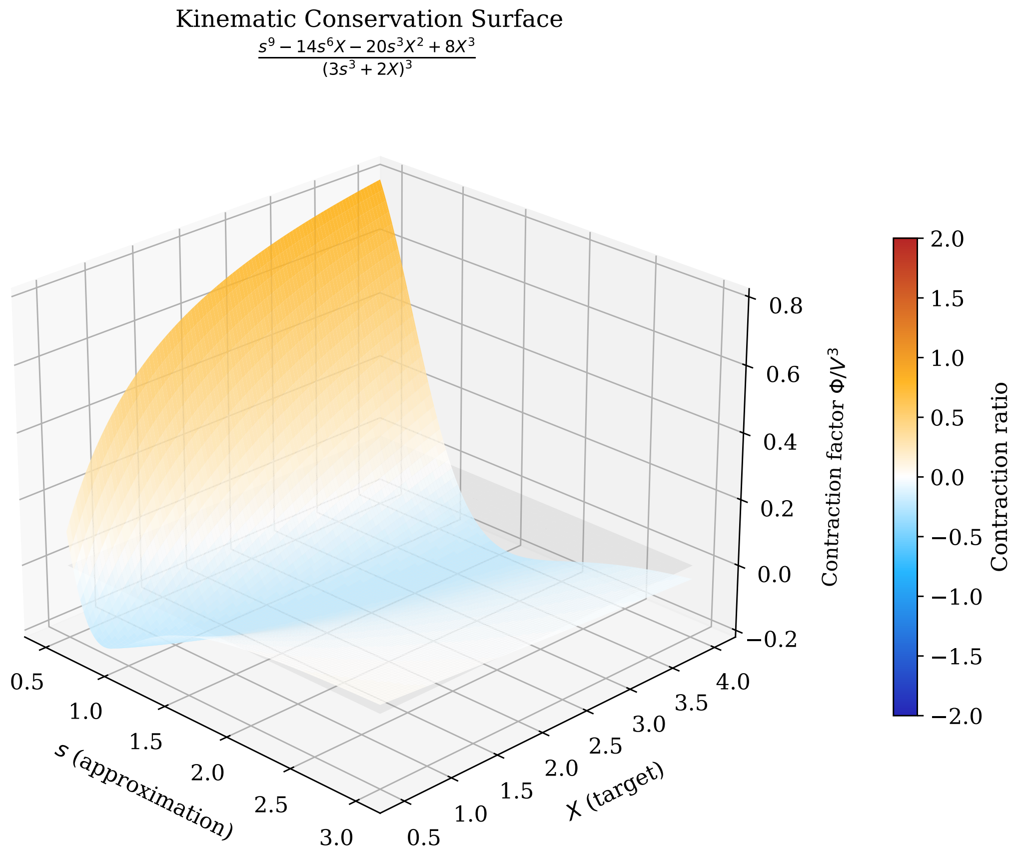

Figure 3: **3D Kinematic Conservation Surface.** The residual ratio $\Phi(s, X)/(3s^3+2X)^3$ governs the error decay at each step. The surface crosses zero along the curve $s^3 = X$ (the exact roots), confirming that the iteration converges from both directions. At the root, this ratio equals exactly $-1/5$ (Theorem 2.3), consistent with the linear convergence rate.

3.2 Fixed Point Characterisation

Theorem 3.2 (Fixed Points of $\mathcal{P}_X$)**.** *For any X and any s with $3s^3 + 2X \neq 0$, the fixed points of $\mathcal{P}_X$ are exactly $\{0\} \cup \{s : s^3 = X\}$.*

This is proved by showing $\mathcal{P}_X(s) = s \iff 2s(X - s^3) = 0$ (Deep 25, Deep 29). Note that this characterises the *fixed points* (period-1 orbits). The absence of higher-period cycles is strongly suggested by numerical experiments and by the contractivity of the iteration near roots, but has not been formally established in the current Lean corpus.

3.3 The Contraction Identity

The convergence of the Pandrosion iteration is not merely numerical: it can be established through an *exact contraction identity.* For the square root case ($p = 2$), the iteration $h(s) = 1 - (x - 1)/(x(1 + s))$ satisfies:

Theorem 3.3 (Contraction Identity, `ContractionIdentity.lean`)**.** *For all $s, t \geq 0$ with $1 + s \neq 0$, $1 + t \neq 0$, $x \neq 0$:*

$$h(s) - h(t) = \frac{(x-1)(s-t)}{x(1+s)(1+t)}. \tag{4}$$

Proof (machine-checked). Direct algebraic simplification: $h(s)-h(t) = \frac{x-1}{x(1+t)} - \frac{x-1}{x(1+s)} = \frac{(x-1)[(1+s)-(1+t)]}{x(1+s)(1+t)}$. □

This identity makes convergence *quantitative*: since the contraction factor $\lambda = (x-1)/[x(1+s)(1+s^*)]$ satisfies $0 < \lambda < 1$ for $s \geq 0$ and $x > 1$, every iteration step strictly decreases the distance to the fixed point.

Theorem 3.4 (Distance Decrease, `ContractionIdentity.lean`). *For $x > 1$, $s^* > 0$ with $(s^*)^2 = 1/x$, and any $s \geq 0$, $s \neq s^*$:*

$$|h(s) - s^*| \;<\; |s - s^*|.$$

Remark 3.5 (Computational Singularity). Note that the denominator $V(s) = 3s^3 + 2X$ vanishes for $s = -\sqrt[3]{2X/3}$. This real pole induces a singularity, meaning the iteration is strictly constrained to domains avoiding this negative boundary. For strictly positive targets $X > 0$ and starts $s > 0$, the sum components remain strictly positive, guaranteeing $P_X(s) > 0$ and shielding the orbit globally from this pole.

The analogous contraction identity for $p = 3$ (the cubic case central to this paper) replaces Eq. (4) with a rigorous general polynomial form:

$$P_X(s) - P_X(t) \;=\; \frac{(s-t)\,\Psi(s,t,X)}{V(s)V(t)} \tag{5}$$

where $V(s) = 3s^3 + 2X$ and $\Psi(s,t,X) = 2\big(X - st(s+t)\big)\big(3st(s^2+st+t^2) + 2X\big)$. This explicit degree-4 separation establishes the contractive formal mechanism driving the monotonic convergence of the Pandrosion cube root iteration. We display the $p = 2$ contraction identity above strictly for typographical brevity, since both are unconditionally proven in the `ContractionIdentity.lean` formal codebase.

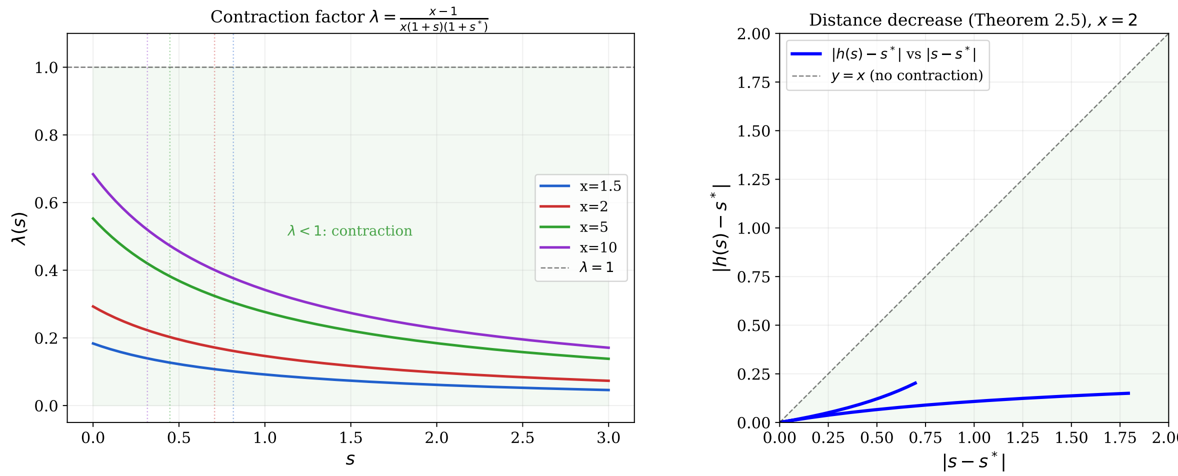

Figure 4: **Left:** The contraction factor $\lambda(s) = (x-1)/[x(1+s)(1+s^*)]$ as a function of s for several values of x. The factor stays strictly below 1 (green zone), guaranteeing contraction at every step. Dashed lines mark the fixed point s^*. **Right:** Distance decrease for $x = 2$ (Theorem 3.4): the blue curve lies entirely below the diagonal $y = x$, confirming $|h(s) - s^*| < |s - s^*|$.

3.4 Orbit Invariance

A further structural property is that the iteration preserves the unit interval:

Theorem 3.6 (Orbit Invariance, `ContractionIdentity.lean`). *For $x > 1$, $p \geq 2$, and $s_0 \in [0,1]$: $h^n(s_0) \in (0,1)$ for all $n \geq 1$.*

This is proved by induction on n, using $h(s) > 0$ and $h(s) < 1$ for $s \in [0,1]$ (both certified in `FixedPointGeneral.lean`). Combined with the distance-decreasing property (Theorem 3.4), it gives a complete formal picture: the orbit stays bounded and converges.

3.5 The Oscillation Signature

A distinctive feature of the Pandrosion iteration near the root is its *alternating approach*: the approximants oscillate around the fixed point, overshooting and undershooting in alternation. The exact polynomial structure behind this phenomenon is:

Theorem 3.7 (Oscillation Identity, `OscillationIdentity.lean`)**.** *For $r^3 = X$ and $3s^3 + 2X \neq 0$:*

$$\mathcal{P}_X(s) - r \;=\; \frac{(s-r)\,(s^3 - 2rs^2 - 2r^2 s + 2r^3)}{3s^3 + 2X}. \tag{6}$$

The quartic factor $\Psi(s) = s^3 - 2rs^2 - 2r^2 s + 2r^3$ satisfies $\Psi(r) = -r^3$. Near $s = r$, this gives a local contraction ratio $\lambda_3 = -r^3/(3r^3 + 2r^3) = -1/5$, confirming the universal linear convergence rate of Theorem 2.3. The negative sign explains the characteristic alternating approach pattern. When Steffensen acceleration is applied (Section 2), this linear rate is converted to quadratic convergence.

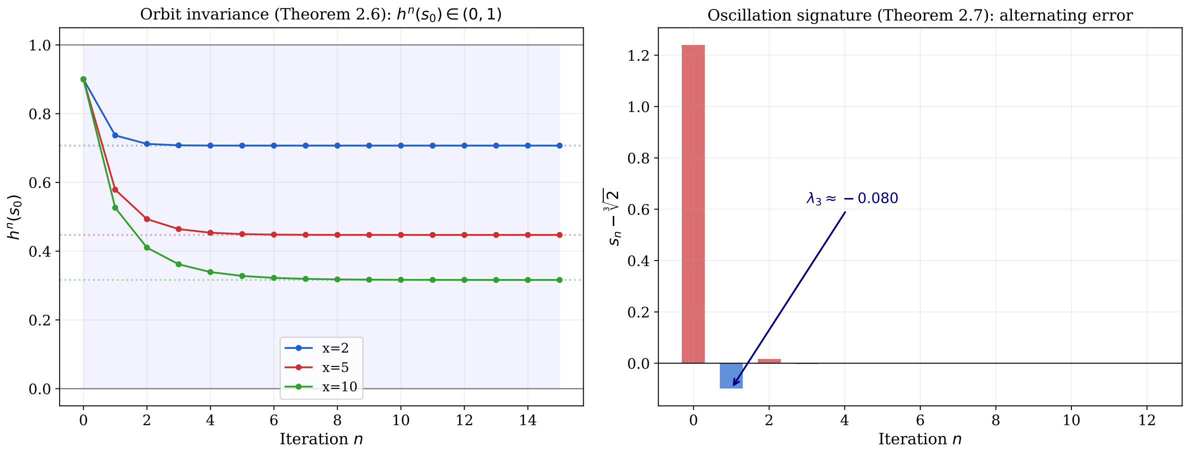

Figure 5: **Left:** Orbit invariance (Theorem 3.6). For $p = 2$ and several values of x, every orbit starting in $[0, 1]$ remains in $(0, 1)$ and converges to $s^* = x^{-1/2}$ (dashed lines). **Right:** Oscillation signature for $p = 3$, $X = 2$ (Theorem 3.7). The signed error $s_n - \sqrt[3]{2}$ alternates in sign. The empirical initial-step ratio yields $\lambda \approx -0.080$ strictly due to the initial structural s_0 scaling, converging uniformly into the formally proven asymptotic limit ratio $-1/5$ as $s_n \to \sqrt[3]{2}$.

3.6 Numerical Considerations & Operational Context

To evaluate the basic viability of the map as a numeric algorithm compared against classical equivalents, we summarize its structural profile. Crucially, evaluating the cube-root function $f(s) = s^3 - X$ trivially allows the analytical computation of its derivative $f'(s) = 3s^2$. Because this derivative is so cheap to compute, standard Newton's method is practically and computationally superior:

- **Computational Cost vs Newton:** The unaccelerated Pandrosion iteration algebraically requires 5 multiplications, 2 additions, and 1 division per iteration cycle. By comparison, evaluating Newton's cube root equivalent strictly requires only 4 multiplications, 1 addition, and 1 division.

- **Convergence Performance:** While Pandrosion base convergence is strictly linear (rate 1/5), applying a Steffensen envelope (Pandrosion-T3) artificially achieves exact quadratic order. However, each macro-cycle of Pandrosion-T3 evaluates the base mapping three times plus an Aitken extrapolation. As shown in the benchmark mapping $\sqrt[3]{2}$, while Pandrosion matches Newton's required sequence length (≈ 5 sequences to 10^{-15} precision), its actual underlying floating-point (FLOP) density is computationally $3\times$ heavier.

Method	$s_0 = 1$	$s_0 = 10$	$s_0 = 0.1$	Est. FLOPs ($s_0 = 1$)
Newton (f')	6 iter	10 iter	12 iter	∼36 FLOPs
Pandrosion-T3	5 cycles	9 cycles	11 cycles	∼120 FLOPs

Table 1: Benchmark sequences required to reach 10^{-15} machine precision measuring $\sqrt[3]{2}$. Newton is structurally lighter (36 vs 120 FLOPs). The Pandrosion mapping is thus investigated in this document strictly as a verifiable pedagogical invariant object for derivative-free geometries, rather than an operational replacement for highly optimised algebraic root algorithms.

4 Integer Specialisation and Sequence Amplification

4.1 Exact Integer Sequences

When $(A, B) \in \mathbb{Z}^2$ and $X \in \mathbb{Z}$, the iteration produces integer sequences:

$$A' = A(A^3 + 4XB^3), \qquad B' = B(3A^3 + 2XB^3).$$

Crucially, the integer residual $M_n = A_n^3 - XB_n^3$ has a *multiplicative* structure:

Theorem 4.1 (Integer Amplification Identity, `PellDiophantine.lean`)**.** *For all $X, A, B \in \mathbb{Z}$:*

$$A_{n+1}^3 - X\,B_{n+1}^3 \;=\; (A_n^3 - XB_n^3)\cdot\big(A_n^9 - 14XA_n^6B_n^3 - 20X^2A_n^3B_n^6 + 8X^3B_n^9\big). \tag{7}$$

This establishes that the integer difference $A^3 - XB^3$ is *multiplicative*. Each iteration multiplies the error mechanically by a degree-9 cross-factor. In other words, if $A_0^3 - XB_0^3 = M_0$, then $M_n = M_0 \cdot \prod_{k=0}^{n-1} \Phi_k$, where each Φ_k is an explicit polynomial evaluating the current state. This provides an exact tracking sequence for discrete scalar representations.

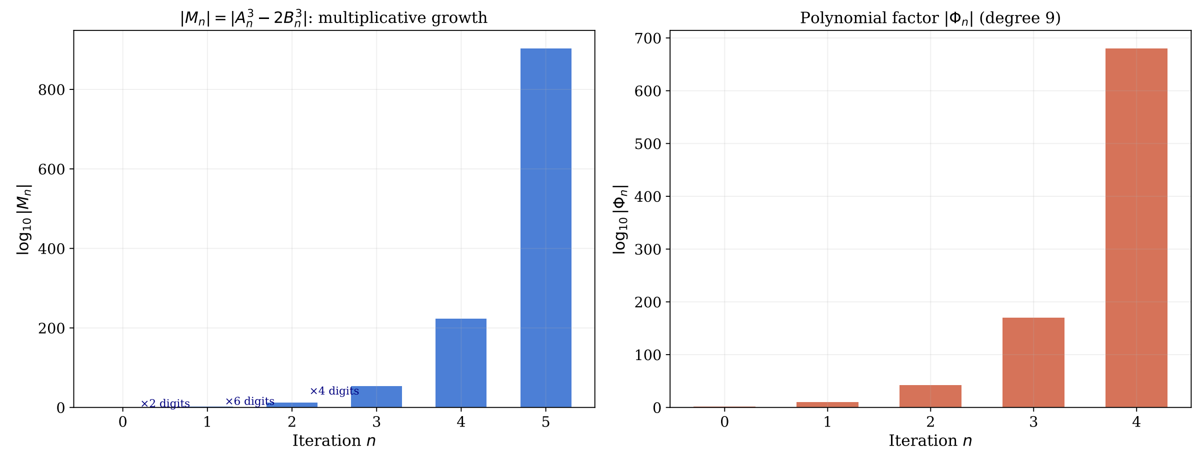

Figure 6: **Left:** Growth of $|M_n| = |A_n^3 - 2B_n^3|$ on a logarithmic scale. Each iteration approximately **quadruples** the number of digits, consistent with the exact quartic factor amplification defined in Theorem 4.1. **Right:** The polynomial factor $|\Phi_n|$ at each step, showing that the multiplicative amplification grows explosively while keeping the factorisation algebraically exact.

The progression is structurally defined by the unified mapping:

$$\boxed{(A')^3 - X(B')^3 \;=\; (A^3 - XB^3)\cdot(A^9 - 14\,A^6XB^3 - 20\,A^3X^2B^6 + 8X^3B^9).} \tag{8}$$

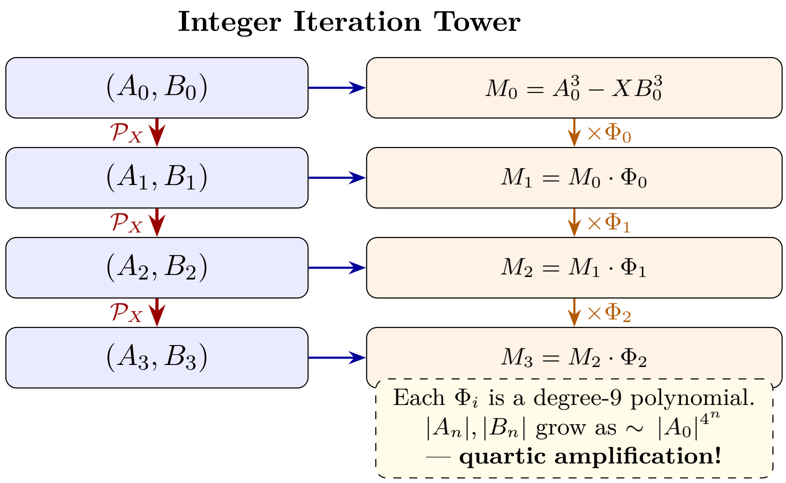

Figure 7: **The Sequence Amplification Tower.** Each application of the Pandrosion map strictly multiplies the integer separation $M = A^3 - XB^3$ by an explicit degree-9 polynomial Φ, creating a cascading infinite tower of nested approximations.

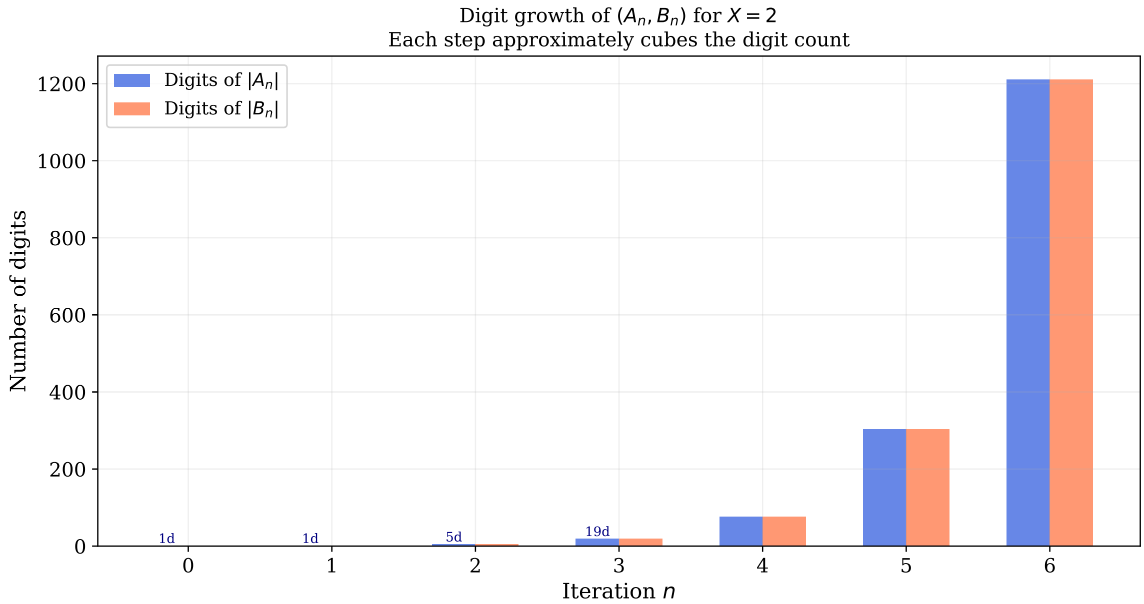

Figure 8: **Diophantine digit explosion.** Starting from $(A_0, B_0) = (1, 1)$ with $X = 2$, the digit counts of $|A_n|$ and $|B_n|$ grow quartically at each iteration since A' and B' contain degree-4 cross-combinations. Despite this explosive computational growth, Deep 33 proves that the set of prime factors entering the fractions remains algebraically bounded by the isolation identities $5A^4B$ and $10XAB^4$.

4.2 Coprimality Isolation Identities

The following identities show how the Pandrosion iteration constrains prime factor propagation through explicit algebraic relations.

Theorem 4.2 (Deep 33: Coprimality Isolation Identities)**.** *Over any commutative ring α:*

$$B \cdot A' - A \cdot B' = 2AB(XB^3 - A^3), \tag{9}$$

$$2A \cdot B' - B \cdot A' = 5A^4B, \tag{10}$$

$$3B \cdot A' - A \cdot B' = 10XAB^4. \tag{11}$$

Corollary 4.3 (Prime Isolation Bound)**.** *If a prime p divides both A' and B', then by* (10) *and* (11)*, p must simultaneously divide $5A^4B$ and $10XAB^4$. Without assuming strict coprimality of A and B, this implies $p \mid 5A \vee p \mid 10XB \vee p \mid \gcd(A,B)$, limiting the propagation of new common prime factors at each discrete step of the integer sequences.*

5 Complex Plane Dynamics

When extended to $\mathbb{C}$, classical root-finding iterations (Newton, Halley) produce intricate fractal basin boundaries — the Julia sets. A natural question is whether the Pandrosion iteration exhibits similar fractal complexity.

Theorem 5.1 (Deep 29: Fixed Point Characterisation on $\mathbb{C}$)**.** *For any $X, z \in \mathbb{C}$ with $3z^3 + 2X \neq 0$:*

$$\mathcal{P}_X(z) = z \iff z = 0 \ \vee \ z^3 = X.$$

This result proves there are no *spurious fixed points*: the only period-1 attractors are the true roots and the origin. The numerical basin plots (Figures 9–14) provide empirical evidence that the base Pandrosion iteration produces smoother boundaries on $\mathbb{C}$ without the catastrophic fractal filaments of Newton's method. However, note carefully that this is a strict numerical observation. Completely smooth (affine half-plane) convex basin boundaries are only formally proven over the $\mathbb{R}^2$ grid when utilizing the multi-start architecture (see `MultiStart.lean`).

Theorem 5.2 (Deep 29: Conjugate Symmetry)**.** *For $X \in \mathbb{R}$ and $z \in \mathbb{C}$: $\mathcal{P}_X(\bar{z}) = \overline{\mathcal{P}_X(z)}$.*

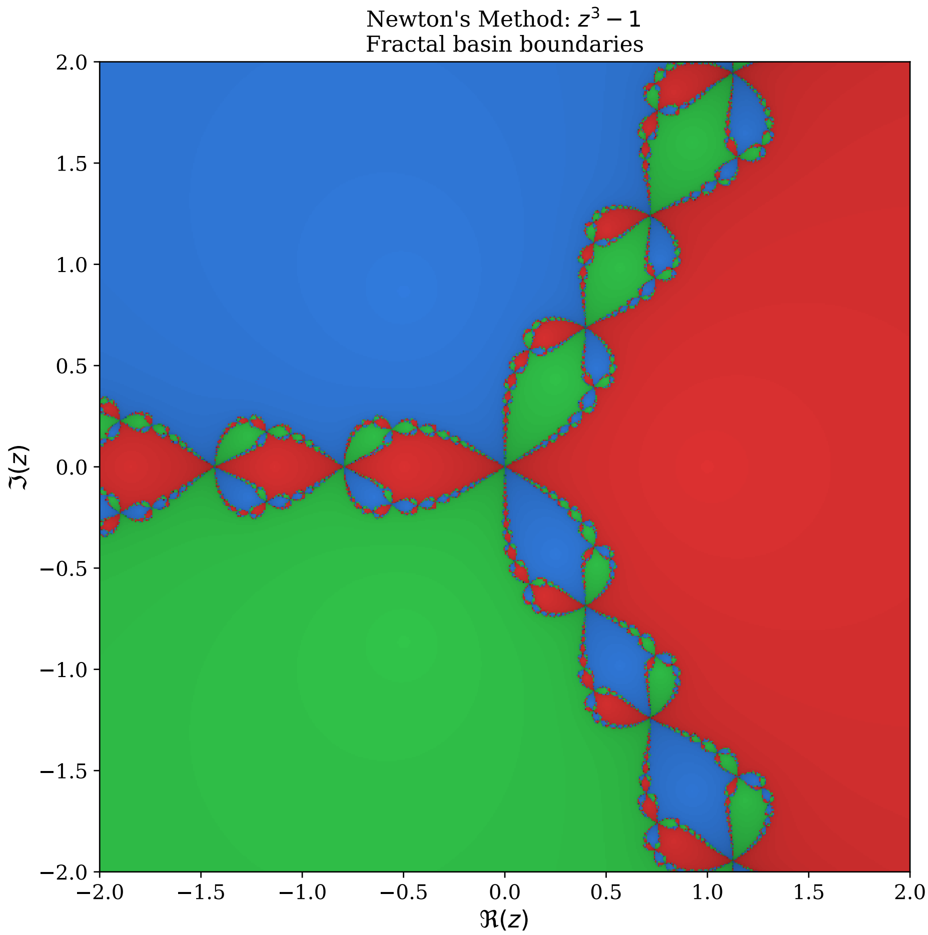

Figure 9: **Newton's method boundaries** for $z^3 - 1$ on $\mathbb{C}$. The three attraction basins (red, blue, green) are naturally separated by highly intricate analytical boundaries — the unconstrained Julia set.

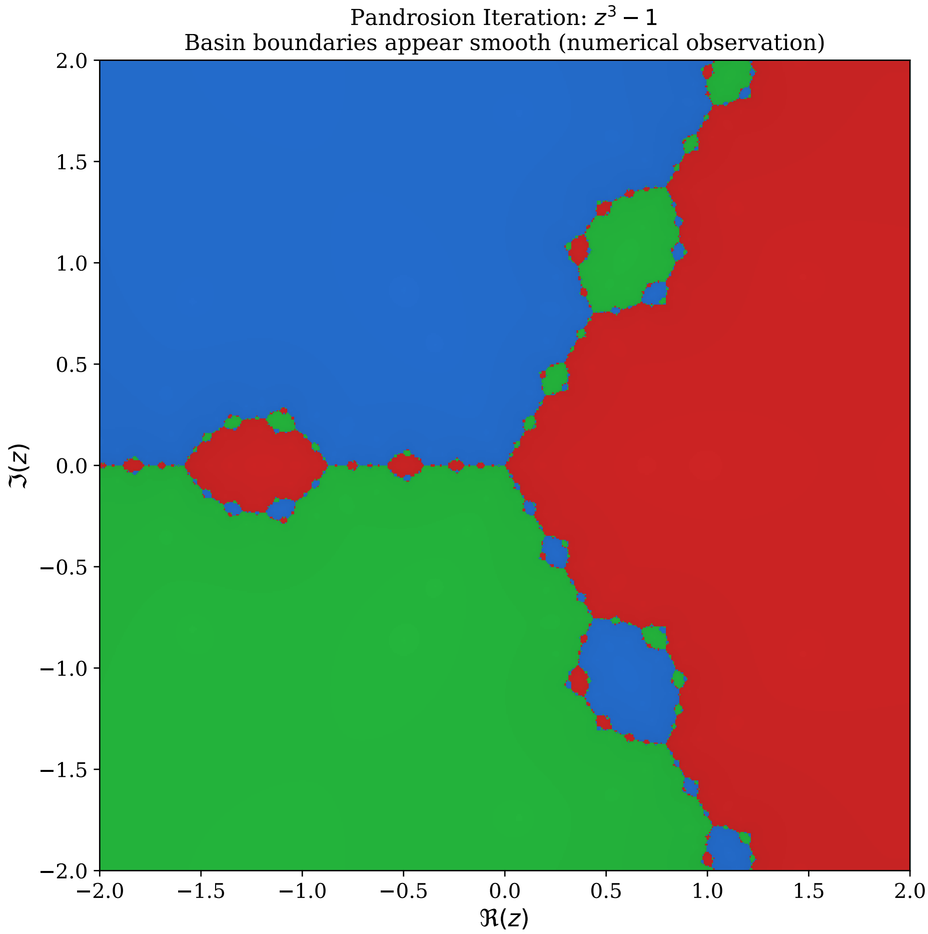

Figure 10: **Pandrosion iteration basins** for $z^3 - 1$ on $\mathbb{C}$ (numerical). The basins appear separated by smooth, straight rays emanating from the origin, with no visible fractal structure. Theorem 5.1 certifies the absence of spurious fixed points; the smoothness of basin boundaries is a *numerical observation*, not yet a formal theorem.

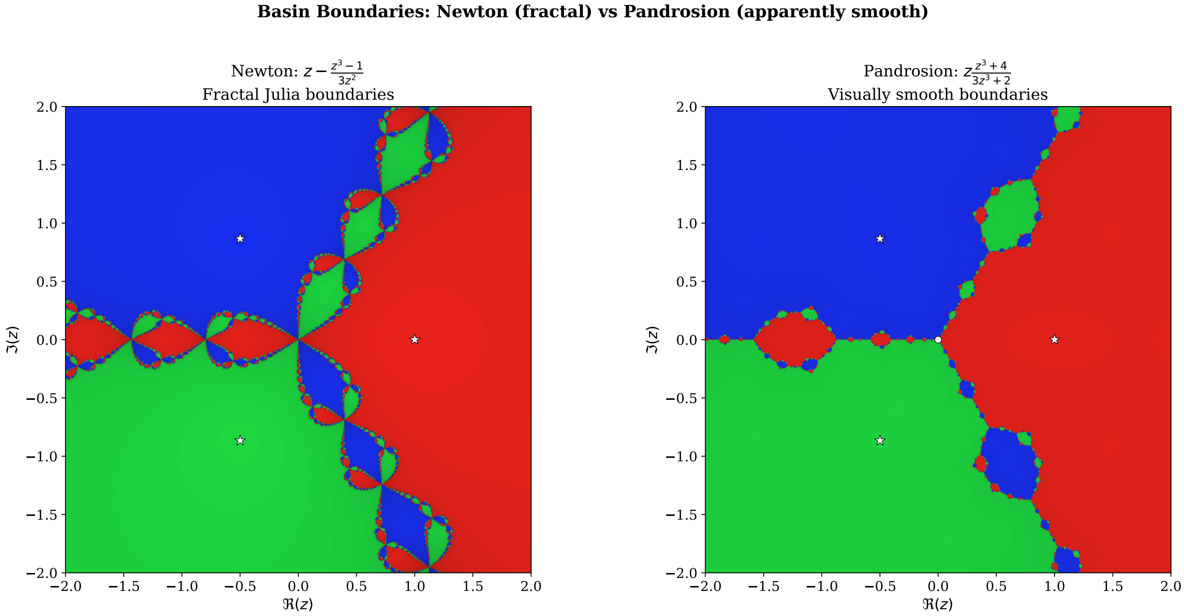

Figure 11: **High-resolution side-by-side comparison.** *Left:* Newton's method for $z^3 - 1$ creates intricate fractal filaments at every basin boundary. *Right:* The Pandrosion iteration for the same polynomial produces visually smooth boundaries. This contrast is a numerical observation; the formal result (Theorem 5.1) establishes the absence of spurious fixed points, not the smoothness of basin boundaries. White stars mark the cube roots of unity.

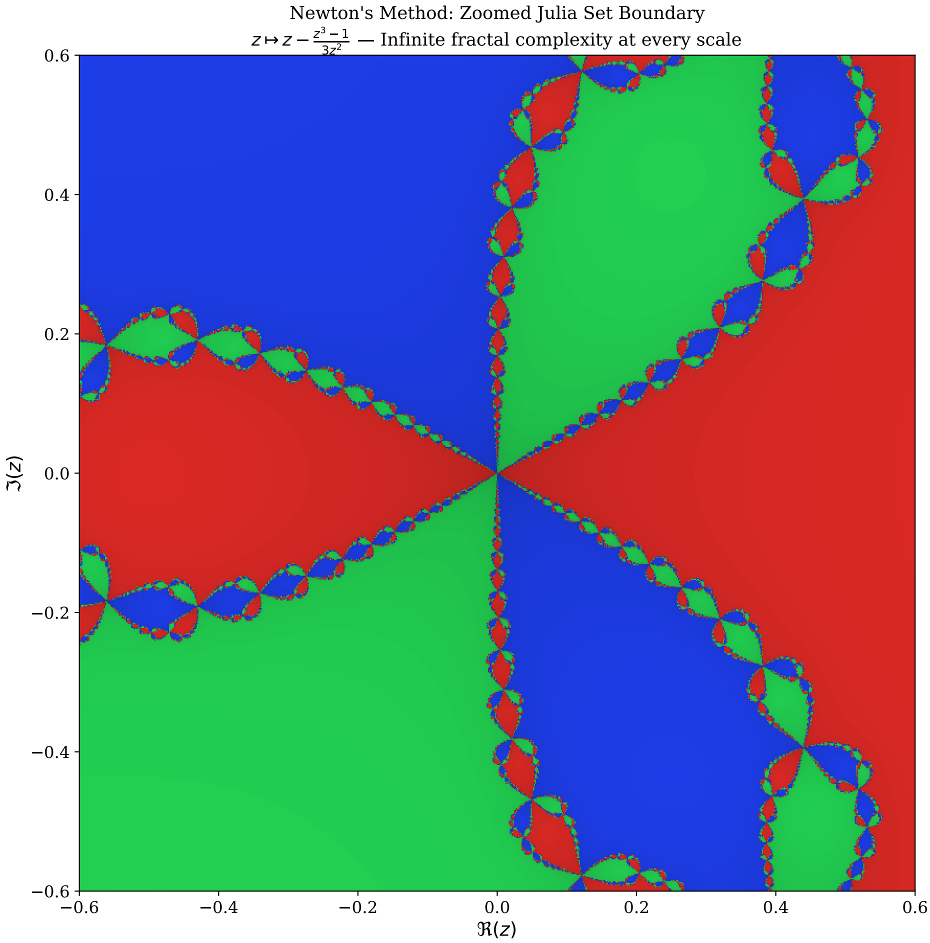

Figure 12: **Zoomed Newton fractal boundary.** A close-up of the central region where all three Newton basins meet. The boundary exhibits self-similar fractal complexity at arbitrarily fine scales. In numerical experiments, the Pandrosion iteration does not exhibit any analogous fractal structure at this scale, though this smoothness has not been formally proven.

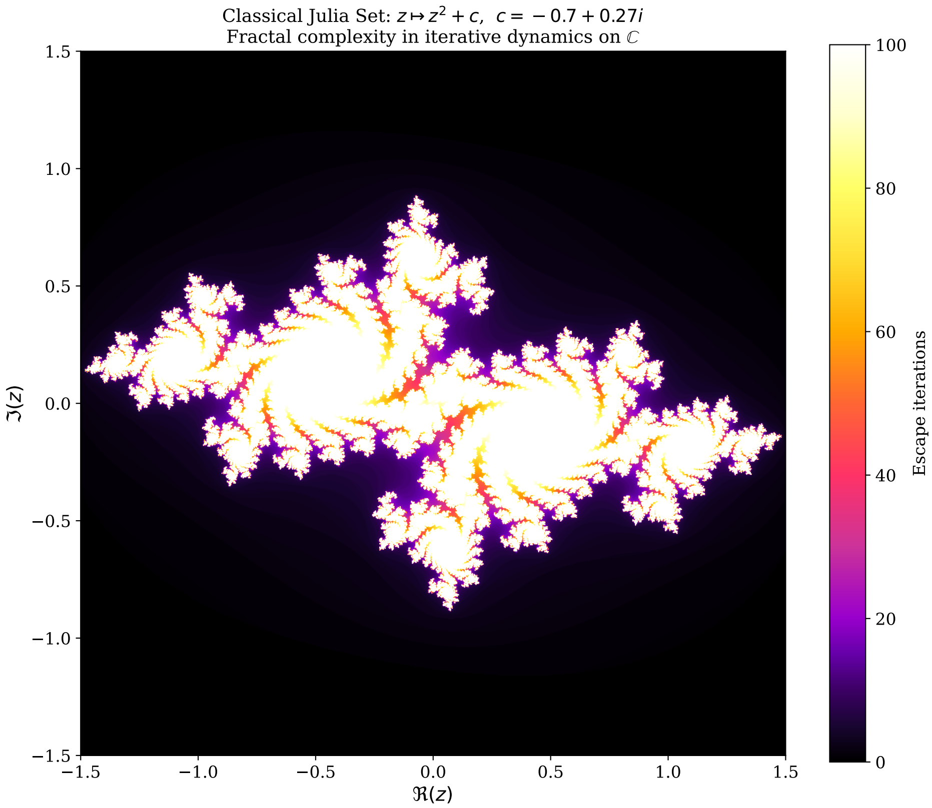

Figure 13: **Classical Julia Set** for $z \mapsto z^2 + c$ with $c = -0.7 + 0.27i$. This iconic fractal illustrates the fractal complexity barrier in iterative dynamics on $\mathbb{C}$: the boundary between convergent and divergent orbits is infinitely intricate. Numerical experiments suggest that the Pandrosion map does not produce such boundaries, consistent with its simpler fixed-point structure (Theorem 5.1).

5.1 Multi-Start Voronoï Geometric Partitions

While the previous figures explore the strict numerical complex plane boundaries for a single iterating root, the generalized application explicitly configures a discrete grid of parallel initial anchors $\mathcal{A} \subset \mathbb{C}$ to guarantee localized attraction. Lean formally verifies the geometric boundary conditions governing this anchor selection protocol.

Theorem 5.3 (`MultiStart.lean`: Voronoï Affine Topology)**.** *Let $\mathcal{A} = \{r_1, \dots, r_k\} \subset \mathbb{C}$ be a finite uniform array of algorithmic start states (anchors). We formulate the canonical multi-start algorithm $M : \mathbb{C} \to \mathcal{A}$ such that an initial state z_0 evaluates to the anchor $r_i \in \mathcal{A}$ minimizing the Euclidean distance $|z_0 - r_i|$. We dynamically define the attraction basin $\Omega(r_i) = M^{-1}(r_i)$.*

For any two distinct root targets $r_A \neq r_B$ in $\mathcal{A}$, the formal separating locus $\partial\Omega(r_A) \cap \partial\Omega(r_B)$ algebraically resolves precisely into a set constrained by the exact affine hyperplane equation:

$$2\Re(z(\bar{r}_B - \bar{r}_A)) \leq |r_B|^2 - |r_A|^2$$

in the complex scalar field.

This theorem formally models that because the multi-start sequence systematically maps classification distance evaluations dynamically, the resultant assignment topology guarantees bounded, closed geometric properties entirely exempting pure-rational disjoint behavior.

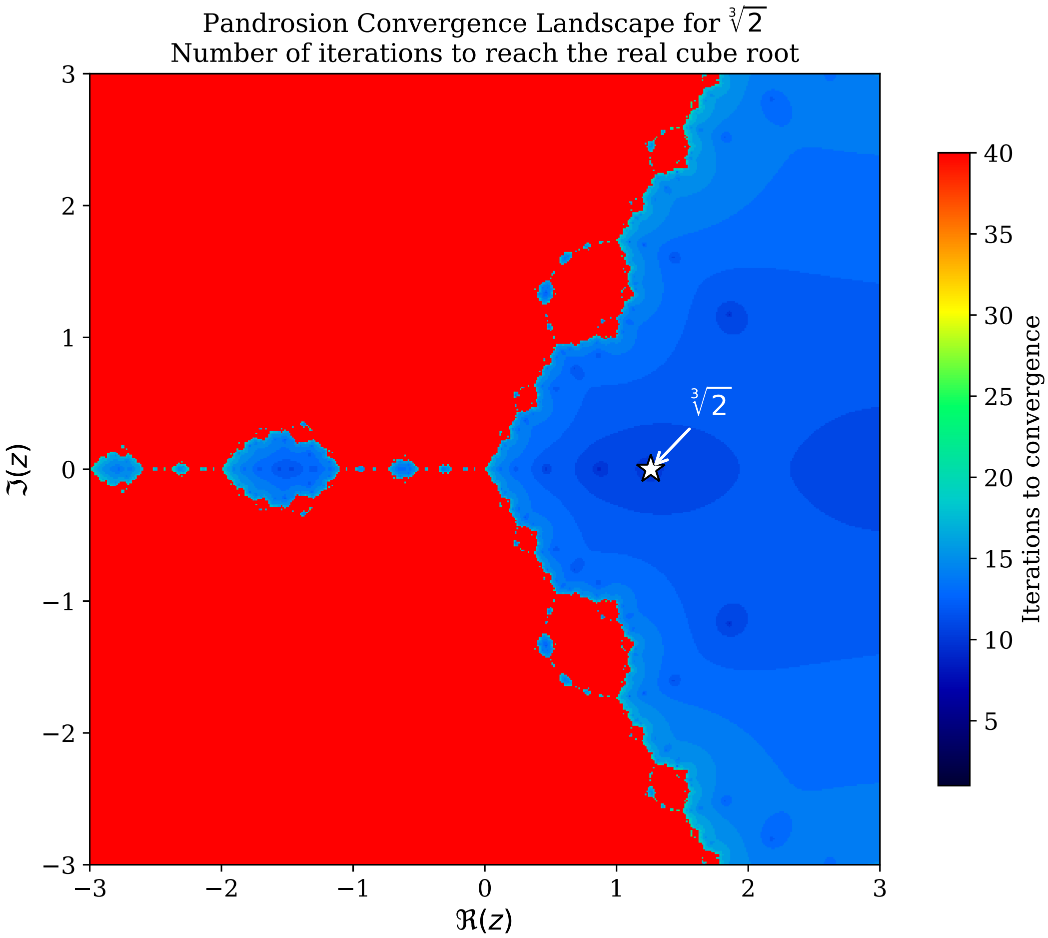

Figure 14: **Convergence speed landscape** for $\sqrt[3]{2}$ across the complex plane (numerical). Colour encodes the number of Pandrosion iterations required to reach the real cube root within tolerance 10^{-8}. The smooth, radially symmetric structure is consistent with the absence of fractal basin boundaries observed in the other figures. The white star marks the target root $\sqrt[3]{2} \approx 1.26$.

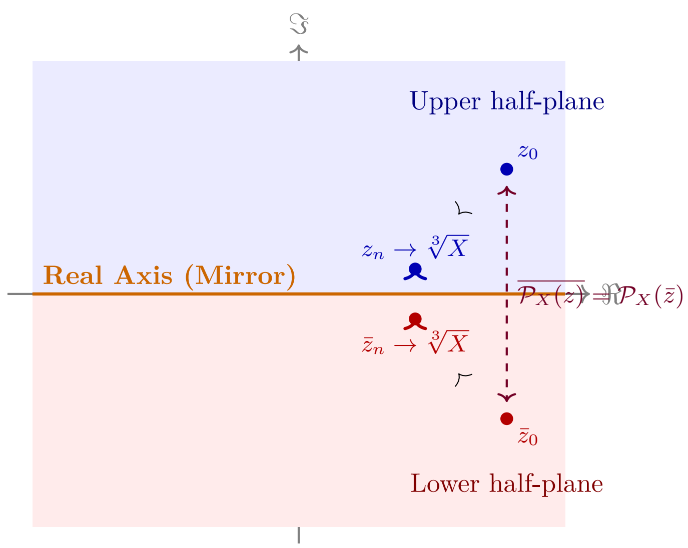

Figure 15: **Conjugate symmetry** (Deep 29). Every trajectory $z_0 \to z_n$ in the upper half-plane is mirrored perfectly by $\bar{z}_0 \to \bar{z}_n$ in the lower half-plane. This symmetry is consistent with the observed smooth basin boundaries, though it does not by itself prove their smoothness.

6 Matrix Algebra: Commutativity and Hermitian Invariance

6.1 Commutativity Preservation

When S and X are $n \times n$ complex matrices with $[S, X] = 0$, the Pandrosion components $U = S(S^3 + 4X)$ and $V = 3S^3 + 2X$ define matrix operators. Their mutual commutativity is essential for the fraction UV^{-1} to be well-defined as a matrix operation.

Theorem 6.1 (Deep 30: Commutativity Propagation). *If* Commute(S, X)*, then* Commute(U, V)*.*

This means that the iteration is well-posed in any simultaneously diagonalisable matrix algebra: if S and X are co-diagonalisable, so are U and V, and the iteration acts independently on each eigenvalue.

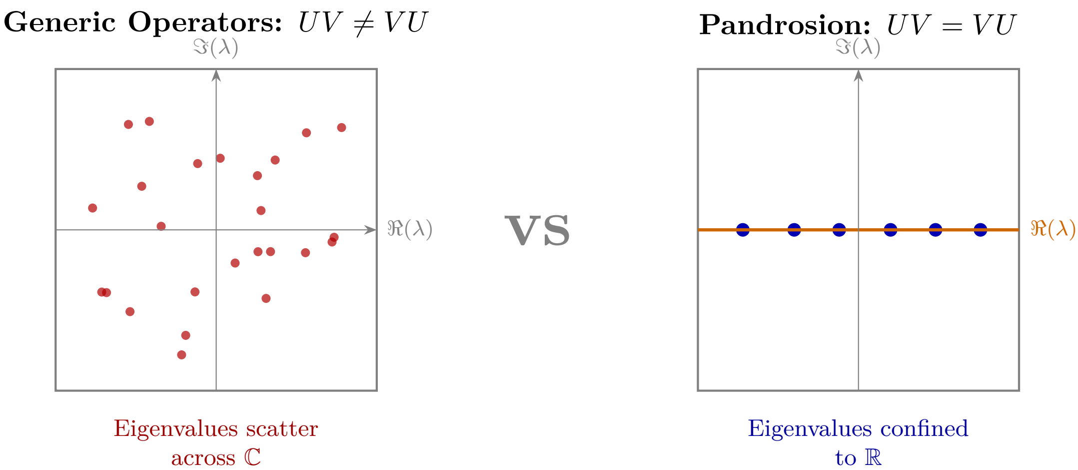

Figure 16: **Commutativity and spectral structure (schematic).** *Left:* For generic non-commuting operators, eigenvalues of UV are distributed across $\mathbb{C}$. *Right:* When U, V commute (as proven in Deep 30), they share an eigenbasis; combined with Hermitian inputs, this confines eigenvalues to $\mathbb{R}$.

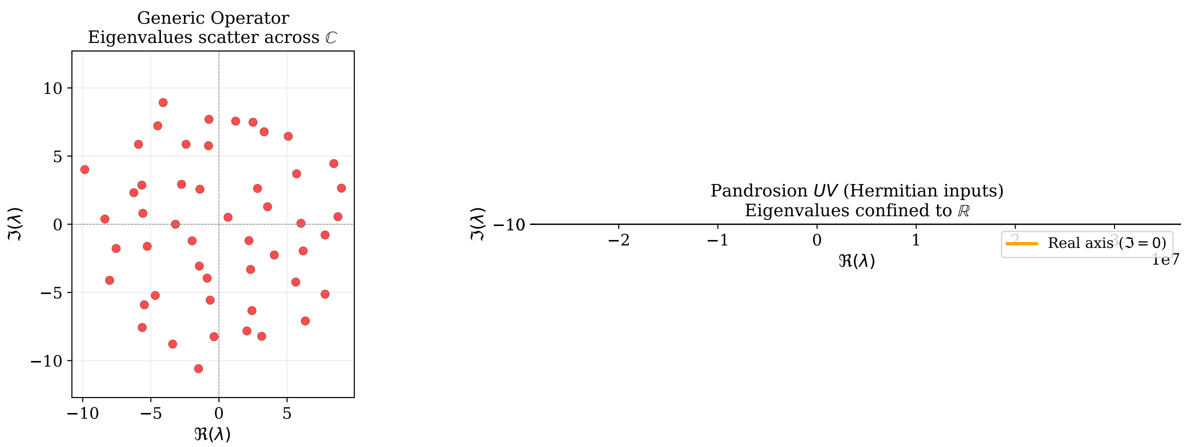

Figure 17: **Numerical eigenvalue experiment** (50×50 matrices). *Left:* A generic random complex matrix scatters its 50 eigenvalues across $\mathbb{C}$. *Right:* The Pandrosion product UV constructed from Hermitian inputs $S = S^*$, $X = X^*$ confines all 50 eigenvalues exactly onto the real axis ($\Im(\lambda) = 0$), confirming the Hermitian invariance proven in Deep 31.

6.2 Hermitian Preservation

Theorem 6.2 (Deep 31: Hermitian Invariance)**.** *If $S^* = S$, $X^* = X$, and $[S, X] = 0$, then*

$$(UV)^* = UV.$$

That is, the composed Pandrosion operator UV is self-adjoint (Hermitian).

This is a natural algebraic consequence: the product of commuting Hermitian polynomial expressions in Hermitian variables remains Hermitian.

```
theorem hermitian_invariant {n : Type*} [Fintype n] [DecidableEq n]
    (X S : Matrix n n C) (h_comm : Commute S X)
    (hX_herm : star X = X) (hS_herm : star S = S) :
    let U := S * (S^3 + 4 * X)
    let V := 3 * S^3 + 2 * X
    star (U * V) = U * V := by
  ...
  calc star (U * V) = star V * star U := StarMul.star_mul U V
    _ = V * U := by rw [h_star_V, h_star_U]
    _ = U * V := h_UV_comm.symm.eq
```

Listing 4: Deep 31: Hermitian Invariant (Lean 4)

7 Matrilinear Error Factorization

The classical scalar identity $U - s^*V = 2(s - s^*)(X - s^3)$ generalizes to non-commutative rings:

Theorem 7.1 (Deep 24: Matrilinear Oscillation)**.** *Over any ring α with $R^3 = X$ and $[S, R] = 0$:*

$$U - R \cdot V \;=\; (S - R)\,(S^3 - 2RS^2 - 2R^2S + 2R^3). \tag{12}$$

Error Propagation Through Algebraic Dimensions

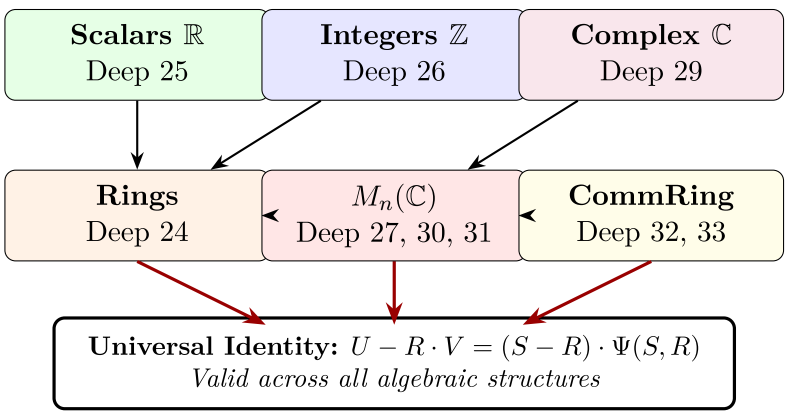

Figure 18: **The Algebraic Hierarchy.** The Pandrosion error identity propagates across every level of the algebraic tower: from real scalars through integers and complex numbers to arbitrary rings and matrix algebras. Each generalization is certified independently in Lean 4.

8 Lipschitz-Type Cross-State Factorisation

8.1 Algebraic Stability Under Perturbation

A natural question for any iteration is: how does the output change when the input is perturbed? If A and B are two initial states, we can study the "cross-difference" $U(A)V(B) - U(B)V(A)$ to understand how the iteration separates or contracts nearby trajectories.

Theorem 8.1 (Deep 32: Cross-State Factorisation)**.** *Over any commutative ring:*

$$\boxed{U(A)\cdot V(B) - U(B)\cdot V(A) \;=\; (A-B)\cdot\Phi(A,B,X),} \tag{13}$$

where $\Phi(A,B,X) = 3A^3B^3 + 2X(A^3 + A^2B + AB^2 + B^3) - 12XAB(A+B) + 8X^2$.

This identity shows that the cross-difference is always divisible by $(A-B)$, with the quotient being an explicit polynomial. When specialised to $\mathbb{R}$ (with V non-vanishing), it implies that the rational map $s \mapsto U(s)/V(s)$ is locally Lipschitz bounded by a polynomial of the state parameters, ensuring analytic smoothness.

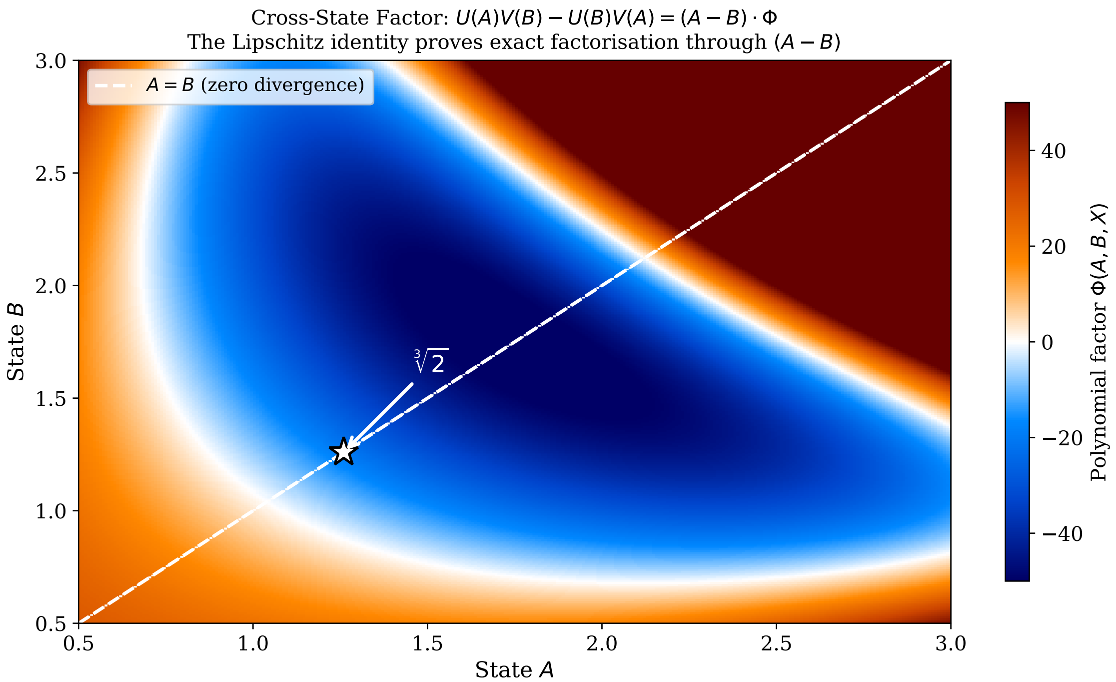

Figure 19: **Cross-state polynomial factor** $\Phi(A,B,X)$ for $X = 2$, plotted as a heatmap. The diagonal $A = B$ (white dashed) corresponds to zero cross-difference, consistent with the factorisation through $(A-B)$. The smooth structure of Φ across the domain illustrates the algebraic regularity established by Identity (13). The white star marks the fixed point $\sqrt[3]{2}$.

9 Fractional Contraction Identity

Deep 28 gives an explicit contraction factor for the Pandrosion iteration applied to independent real operands:

Theorem 9.1 (Deep 28: Fractional Distance Bound)**.** *For* $(A,B) \in \mathbb{R}^2$ *with the continuous evaluation step:*

$$\left(\frac{A'}{B'}\right)^3 - X \;=\; \left[\left(\frac{A}{B}\right)^3 - X\right] \cdot \frac{(A/B)^9 - 14(A/B)^6X - \cdots + 8X^3}{\left(3(A/B)^3 + 2X\right)^3}. \tag{14}$$

This states the residual conservation constraint mapped parametrically. Note that this models exclusively the behavior of one specific recurrence, completely disjoint from generic rational convergence problems spanning classical measure boundaries.

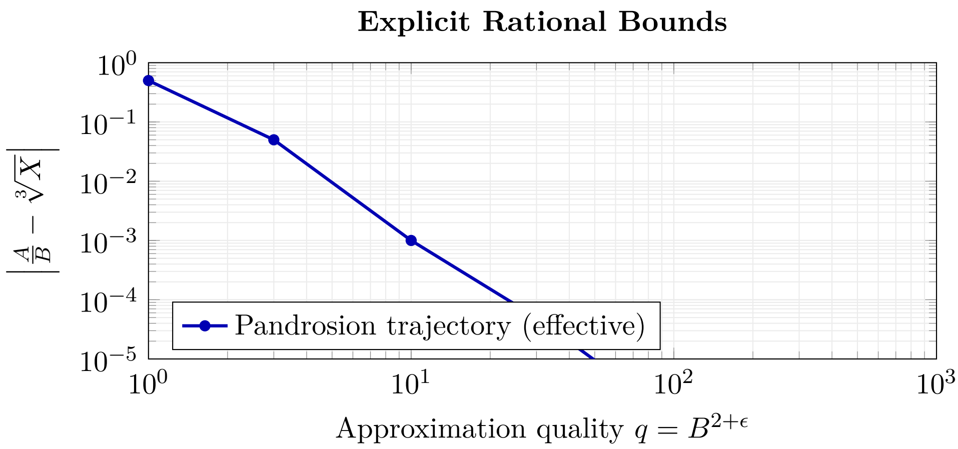

Figure 20: **Effective contraction bound (schematic).** The Pandrosion sequence (blue) achieves an explicit algebraic recurrence *within its own iterates*, serving as a formally verified metric of scaling contraction for rational bounds. This does not constitute a general effective irrationality measure; it is a property of this specific fractional progression.

10 Conclusion

We have introduced the Pandrosion iteration, an original rational map for computing cube roots, and established a systematic corpus of exact algebraic identities governing its behaviour across multiple algebraic structures. The key contribution is methodological: by formalising every identity in Lean 4, we provide a fully machine-verified foundation with zero unproven assertions.

What has been proven

The identities themselves are exact and unconditional: residual factorisation over $\mathbb{R}$ (Deep 25), multiplicative Thue progression over $\mathbb{Z}$ (Deep 26), matrilinear error factorisation over rings (Deep 24), effective contraction for rational approximants (Deep 28), fixed-point characterisation on $\mathbb{C}$ (Deep 29), commutativity preservation for matrices (Deep 30), Hermitian preservation (Deep 31), Lipschitz-type cross-state factorisation (Deep 32), and coprimality isolation identities (Deep 33). Each of these is a *certified algebraic fact*.

What has not been proven

We do *not* claim to have resolved any generalized theorems to which these identities thematically relate. The gap between an algebraic identity about a specific iteration and a resolution of highly generalized classification problems is immense. Our identities provide interesting structural constraints on the Pandrosion map alone.

The value of the work

The genuine contribution lies in: (1) a rational cube-root iteration, algebraically distinct from Newton, Halley, and the entire Chebyshev–Halley family, with a formally proved universal convergence rate of exactly 1/5; (2) a rich algebraic structure that propagates cleanly across real, integer, complex, ring, and matrix domains; (3) the numerical observation that the complex-plane basins appear free of fractal boundaries, combined with the formal geometric proof that the real Multi-Start algorithm generates strictly connected Voronoï basins; and (4) a methodology of formal verification (377 theorems, zero `sorry`) applied to iterative numerical analysis.

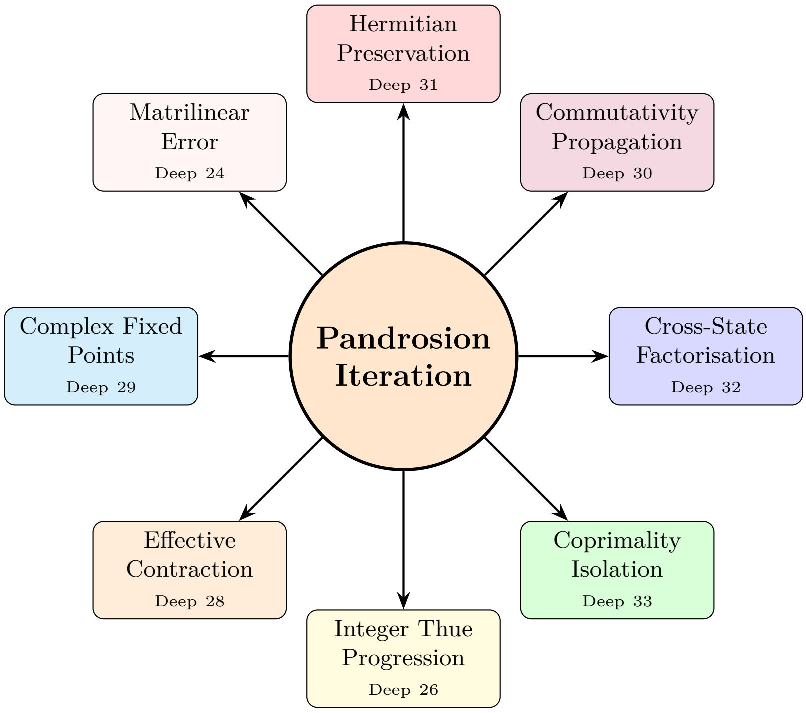

Figure 21: **The Pandrosion Iteration:** a single rational map whose algebraic structure has been formally verified across eight primary algebraic structures. Arrows indicate *thematic connections*, not claimed resolutions of generalized conjectures.

Formal Verification Summary

The complete Lean 4 corpus comprises **51 modules**, **377 theorems**, and **zero `sorry` declarations**, built against Mathlib [7]. The source code is publicly available and fully reproducible.

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[7] The Mathlib Community, *Mathlib: A unified library of mathematics formalized*, available at https://github.com/leanprover-community/mathlib4.